1. Basic Probability Concepts

1.1 Introduction

1.1.1 What is Statistics?

Examples of statistical problems:

- Based on data from a clinical (medical) trial, a researcher hopes to show that a new drug is more effective than the current standard treatment.
- A political pollster would like to predict an election outcome, based on a survey of some likely voters.
- How certain are we that changes in surface temperature over the last four decades can be attributed to human activity?
- An education at Texas A&M is said to provide a good return (e.g., starting salary) on investment (tuition). How is such a conclusion determined?

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What is Statistics?

- *Statistics* is the science of drawing conclusions (inferences) from incomplete or uncertain information. Note that data are limited to what we can actually observe and therefore do not present a complete picture of reality. Almost all data are also subject to uncertainty of one sort or another, even despite the efforts of the researcher.
- A *statistical inference* is any procedure that draws such a conclusion and also *quantifies the inherent uncertainty* of the conclusion. Note that "inference" is *inductive* – reasoning from the specifics of observed data to general statements about reality.
- A *statistical model* is a framework of experimental conditions and procedures, assumptions and mathematical theory that allows for *qualified* statistical inference.
- Statistical models usually involve probability in one or more ways, including as a means for quantifying uncertainty.

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Political Poll

Example 1–1. If 450 out of a sample of 1000 likely voters say they will vote for the Republican candidate, what can we predict about the percentage of *all* voters who intend to vote for the Republican candidate?

Note that the *observed* proportion 45% is almost certainly not the exact proportion that we desire to know.

Here's an example of how the conclusion could be stated in statistical terms:

The proportion of voters who intend to vote for the Republican candidate is between 42% and 48%, with 95% confidence.

The conclusion *quantifies the uncertainty in two ways*:

- By providing a "confidence" level that states how reliable the polling and statistical methods are.
- By giving an interval of plausible values for the proportion that reflects the uncertainty due to making a conclusion based on only 1000 voters.

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Populations and Samples

Most statistical problems can be phrased in terms of the following:

- A *population* is the *entire collection* of individuals about which one wants to draw a conclusion or make an inference. Populations can be fixed or conceptual, finite or infinite.
- A *sample* is the *observed subset* of the population, which typically is a very small fraction of the population.

Somewhat surprisingly, if the sample is obtained in an appropriate way, it can provide accurate results about the entire population even though it is relatively quite small.

Example 1–1 (cont.). In our example, the population is the set of all voters while the sample is just the 1000 voters who were polled.

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Induction vs. Deduction

To summarize, inferences about a population based on information in a sample are marked by uncertainty because the sample is only a fraction of the population, and there can be other sources of uncertainty as well. Such conclusions are called *inductive*.

- *Induction* – Reasoning from specific information to general information.
- *Deduction* – Reasoning from general information to specific information.

In our statistical paradigm:

$$\begin{aligned} \text{sample} &\iff \text{specific (observed data)} \\ \text{population} &\iff \text{general (unobserved values)} \end{aligned}$$

The purpose of proper statistical analysis is to manage and report the uncertainty of inferences that are made.

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1.1.2 What is Probability?

- *Probability* is the mathematical tool used for describing and quantifying uncertainty when it is *predictable* in some meaningfully random sense.
- However, probability is not appropriate in all circumstances, such as when looking at haphazard or whimsical behavior.
- Even when used appropriately, its interpretation can be subjective or objective.

Example 1–2. When my doctor says that a treatment works for 50% of patients, he is making an *objective prediction* (for a randomly selected patient). But when I interpret that to mean that I (personally) have a 50% chance of recovery, I am being *subjective*.

Example 1–3. When an engineer says a particular nuclear reactor has a 1 in a million chance of experiencing a meltdown, that is a *subjective* claim in that it refers to a one-time scenario, even if the value of the chance is determined objectively by science and engineering.

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Mere Uncertainty is not Randomness

Suppose I tell you that the month I was born in begins with a "J". You guess January. What is the chance you are correct?

Actually, there is no truly meaningful "chance" for that. My birthdate is not random. And there is no chance applicable to the guess that you *have already made*. You may think you guessed "at random", but it really was only *whimsical* – who's to say that the chance you would choose correctly was $1/3$?

However, suppose you decide that you will select a month (January, June or July) using an random mechanism that gives the same probability to each month. Then, *prior to the selection*, the chance that the selection *will be* my birth month (July) is $1/3$. In this case the choice is random.

Many events that we attribute to randomness really only occur as a result of someone/something that is haphazard or whimsical. While we might sometimes try to model them as random, that may very well *require a great deal of assumption*.

Probability is meaningful only in contexts where one can sensibly describe or assume a random mechanism.

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Chaos versus Randomness

- Loosely defined, a *chaos* is a mathematical system that is highly sensitive to initial conditions (the so-called "butterfly effect"). A simple example is the sequence generated recursively by

$$x_{n+1} = 4x_n(1 - x_n), \quad \text{with } 0 < x_n < 1.$$

  Even a very small deviation in the value of x_0 will result in x_{100}, for example, being quite different.

- However, many chaotic systems have statistical properties that emulate random behavior.

- A so-called *pseudo-random number generator* is a well-designed chaos that produces sequences of numbers that (mostly) behave like a random sequence. They are very useful for *simulating* random variables and sequences.

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Probability in Statistics

Probability has multiple uses for statistics.

- Probability can be used to describe a population. Essentially, one is asking about what one would observe for a randomly selected individual in the population. In this sense, *"probability" is equivalent to "proportion"*.
- Methods for randomly selecting an individual, or a sample of individuals, use probability.
- When a random sample is to be obtained, the tools of probability can tell us what to expect or predict about the sample.
- After the fact, once we have the sample in hand and have observed the data, we can quantify the uncertainty that came from sampling and we can compare the data to what was predicted.

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Political Poll (cont.)

Example 1–1 (cont.). For the political poll of 1000 likely voters:

- A randomly selected individual from the population of all voters has a chance p of being someone who intends to vote for the Republican candidate. (If the mechanism for doing the random sampling is *unbiased* then p is the proportion of all voters who intend to vote for the Republican candidate.)
- The pollster uses some random method for selecting individuals. For polls, in particular, it is not a simple lottery – it often is complex, involving first sampling geographical regions, then sampling individuals within the regions, and it also must account for the reluctance of some voters to respond.

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Political Poll (cont.)

Example 1–1 (cont.).

- Having obtained the poll and observed the proportion 45%, and also having set the confidence level (95%), the pollster can compute a range of plausible vales for p (e.g., 42%–48%). For example, let $\widehat{p}$ denote the proportion for the sample. In the simplest scenario for random sampling (a simple lottery), the pollster knows, *prior to obtaining the sample*, that

$$\mathsf{P}\left(|\widehat{p} - p| \leq 2\sqrt{\frac{p(1-p)}{1000}}\right) \approx 0.95.$$

  Estimating the bound with $2\sqrt{\frac{0.45(0.55)}{1000}} = 0.03$, the pollster can then say, with 95% confidence, that p is within 0.03 of 0.45.

  (Political polls are actually much more complex than this, but if done well then the uncertainty often emulates that of the simple case.)

Justifying methods and conclusions such as the above is a principal purpose of probability for statistics.

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What This Course is About

In order to use probability appropriately for statistical analysis, we must *study its theory*. Specifically, we want to:

- Use probability models to describe random variables – the values observed for randomly selected individuals.
- Understand the tools of probability, such as expectation and variance, that enhance descriptions and inform predictions.
- Understand the effects of probability for transformed random variables and for functions of two or more random variables.
- Apply the above to sample statistics – the random variables resulting from summarizing the data of a random sample. (For example, a sample proportion or average.)

Learning the theory of statistical inference will come in the *next* semester.

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1.2 Sample Space

1.2.1 The Formal Setup

We start by formally defining the mathematical terms and concepts that are the basis for all probability.

- An *experiment* is a random process having a set of possible *outcomes* that can be formally/mathematically identified. Exactly *one outcome*, denoted ω ("little omega"), is the random result of the experiment.

  What makes the experiment random is that the observed outcome *cannot be predicted precisely*, even if the experiment is repeated.

- The *sample space* is the set of all possible outcomes of the experiment. The sample space is denoted Ω ("big omega").

- An *event* is a subset of Ω. This includes the empty set $\emptyset$ and Ω itself.

- A sample space is not necessarily a population. It could be the collection of all possible samples (of a certain size) that may be selected from the population. It can be even more general, such as a collection of infinite sequences.

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Rolling Two Dice

Example 1–4. The experiment is to roll two ordinary six-sided dice. We assume the dice are *distinguishable*: one is red and the other is green. An outcome is denoted (i, j), where i is the value on the red die and j is the value on the green die.

Thus, we can formally describe the sample space by the set of possible *ordered pairs*:

$$\Omega = \{(i, j) : i, j \in \{1, \ldots, 6\}\}.$$

Some events:

$A :=$ "Red die is 1 and green is 5" $= \{(1, 5)\}$

$B :=$ "One die is 1 and the other is 5" $= \{(1, 5), (5, 1)\}$

$C :=$ "Total is less than 5" $= \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)\}$

When an event is identified descriptively, as in the examples above, it still needs to be specific enough to ensure it refers to an exact subset of the sample space.

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1.2.2 Understanding Events

Events are subsets, but they are random and we interpret them accordingly. In fact, the more you learn to think of them *both formally* (as sets) and *descriptively* (as random events), the easier it will be to work out problems correctly.

- When we say an event A *occurs*, we mean that the random outcome ω is an element of A. That is,

  $$\text{“}A \text{ occurs”} = \text{“}\omega \in A\text{”}.$$

  (But the experimental result is still *only one outcome* in A, not every one.)

  Note that since one cannot predict for certain what the outcome will be then one cannot generally predict for certain whether A will occur or not.

- Ω always occurs and $\emptyset$ never occurs.

- $A \subset B$ means that the event A is a subset of B. Observe that $A \subset B$ also means “A occurs $\implies$ B occurs”.

- Two sets are equal if they contain each other:

  $$A = B \iff A \subset B \text{ and } B \subset A.$$

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Events are Sets

- The *union* of events A and B, denoted $A \cup B$, is the event containing all the elements that are in either A *or* B or both. "$A \cup B$ occurs" means *at least one* of the two events occurs.
- The *intersection* of A and B, denoted $A \cap B$, is the event containing all elements that are in both A *and* B. "$A \cap B$ occurs" means *both* of the events occur.
- The *complement* of an event A, denoted A^c, is the set of all elements in the sample space that are not in A. "A^c occurs" means A *does not* occur.
- Events A and B are said to be *mutually exclusive* (or *disjoint*) if they have no elements in common. In other words, $A \cap B = \emptyset$. Several sets, say $A_1, \ldots, A_k$, are mutually exclusive if *each distinct pair* A_i and A_j are disjoint.
- A *partition* is a collection of disjoint sets $A_1, \ldots, A_k$ that has union equal to the sample space: $A_1 \cup \cdots \cup A_k = \Omega$.

Become familiar with these definitions and operations, and how to work with them. Also understand how to interpret them as applied to random events.

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Events are Sets

- *DeMorgan's Laws*.

  (i) $A_1^c \cap \cdots \cap A_k^c = (A_1 \cup \cdots \cup A_k)^c$.

  That is, "none of the A_i's occur" is the opposite of "some A_i occurs".

  (ii) $A_1^c \cup \cdots \cup A_k^c = (A_1 \cap \cdots \cap A_k)^c$.

  That is, "some A_i does not occur" is the opposite of "every A_i occurs".

- For convenience, we often use the notation

$$\bigcup_{i=1}^{k} A_i = A_1 \cup \cdots \cup A_k \quad \text{and} \quad \bigcap_{i=1}^{k} A_i = A_1 \cap \cdots \cap A_k.$$

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Countable and Uncountable Sets

- A *finite* set, obviously, is one with a finite number of elements.
- A *countably infinite* set is one whose elements can be put into 1–1 correspondence with all the positive integers $\{1, 2, \ldots\}$. That is, there is a way to list them completely in a sequence.
- *Countable* sets are either finite or countably infinite.
- A set is *uncountable* if it is not countable.

An example of an uncountable set is all the real numbers in the interval $(0, 1)$ (or, in any interval, for that matter).

There is a major distinction between probability models for countable and uncountable sample spaces.

Actually, uncountable sample spaces can be quite complicated. However, for the most part in this course, countable sample spaces will have so-called *discrete* models and sample spaces that are intervals in the real line will have so-called *continuous* models.

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Rolling until a 6 Appears

Example 1–5. Consider rolling a single die until a 6 appears (and then stopping). Outcomes have the form

$$\omega = (x_1, \ldots, x_n), \text{ with } x_n = 6 \text{ and } x_i \in \{1, \ldots, 5\} \text{ for } i < n,\ n \geq 1.$$

The sample space is countably infinite. (Actually, we have simplified the sample space by excluding infinite sequences in which 6 never appears, which otherwise would comprise an event with probability zero).

If we focus only on the number N of times the die is rolled, but not on what happens prior to the 6, then we have a *reduced* sample space

$$\Omega' = \{1, 2, \ldots\} \text{ with } N \text{ as the outcome.}$$

In general, though not always, it is best to use the simplest sample space that has all the information needed for the problem at hand.

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Uncountable Examples

Example 1–6. A bar of concrete is stressed until it breaks. The outcome, compressive strength (in psi), is a positive real value. Even though the possible measurement values are limited, due to rounding, we usually assume the strength can be any real value in some interval $[a, b]$.

However, a and b are not always known, so for modeling purposes the sample space is $\Omega = (0, \infty)$. This is *uncountable*.

Example 1–7. Roll a single die indefinitely (i.e., infinitely many times). Here, outcomes are *infinite sequences*:

$$\omega = (x_1, x_2, x_3, \dots) \text{ with each } x_i \in \{1, \dots, 6\}.$$

Interestingly, the sample space is *uncountable*.

Although infinite sequences may seem unrealistic for experiments, they play very important roles in the theory of probability and its application to statistics.

Observe that the definition of a sample space and its events (subsets) *does not require probability*. Indeed, any sample space can have infinitely many different probability models. We describe those next.

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1.3 Probability Measure

1.3.1 Definition of Probability

We will use the notation $\mathsf{P}(A)$ to denote "the probability that event A occurs." Often, we simply say "the probability of A".

In other words, it represents the chance that the actual outcome will be some element of A.

Observe that probability applies to events, not to outcomes.

Intuitive notions of probability:

- $\mathsf{P}(A)$ represents a degree of belief that event A will happen.
- $\mathsf{P}(A)$ represents the (long-term) proportion of the times A would occur if the experiment were repeated a very large number of times.

As mentioned earlier, interpretations of a probability can be either objective or subjective. Use of a probability for a one-time (non-repeatable) experiment is necessarily subjective, even if it was computed in some objective manner.

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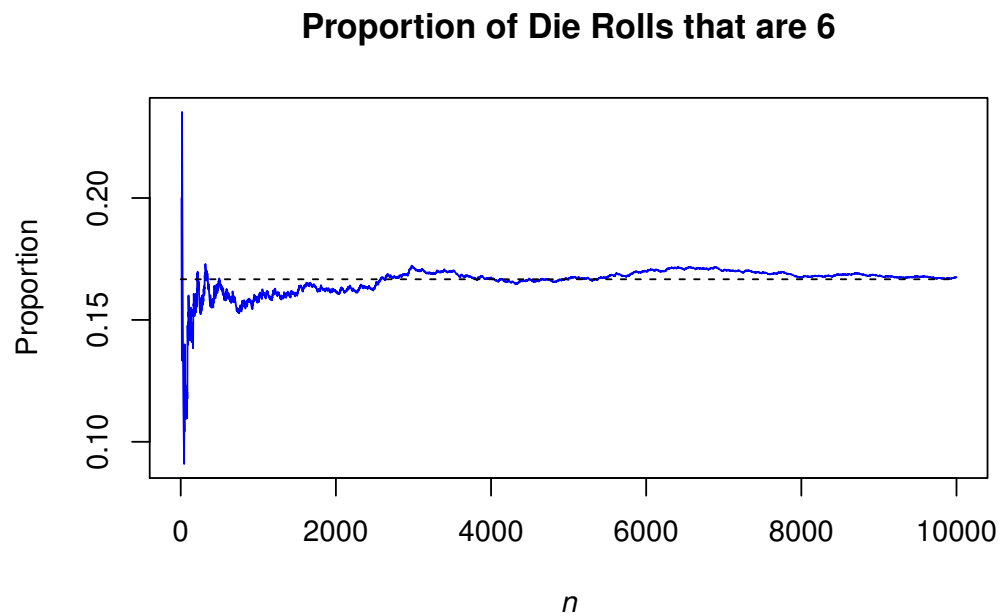
Long-Term Notion

Example 1–7 (cont.). Suppose a fair die is rolled repeatedly (and independently), and we count the number of times that a 6 appears.

We then plot the proportion of times that a 6 appears versus the number of rolls.

The following is a computer simulation of this experiment.

n	$n(6)$	$\frac{n(6)}{n}$
10	2	0.2000
50	7	0.1400
100	16	0.1600
500	83	0.1660
1000	158	0.1580
5000	835	0.1670
10000	1675	0.1675



[longterm.r]

The long-term proportion is (eventually) the same as the chance of a 6 for a single roll. (This actually is a mathematical result.)

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Probability Axioms

A *probability measure* on Ω is a function P from subsets of Ω to the real line satisfying the following three axioms:

Axiom 1 For every event A, $\mathsf{P}(A) \geq 0$.

Axiom 2 $\mathsf{P}(\Omega) = 1$.

Axiom 3 For every infinite sequence of *disjoint* events $A_1, A_2, \ldots,$

$$\mathsf{P}(A_1 \cup A_2 \cup \cdots) = \sum_{i=1}^{\infty} \mathsf{P}(A_i). \quad \textit{(countable additivity)}$$

Here, "axiom" means "minimal required property" for an acceptable definition. However, many interesting and useful properties are logical consequences of just these three axioms.

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1.3.2 Properties of Probability

1. $\mathsf{P}(\emptyset) = 0$.

Proof: $\Omega, \emptyset, \emptyset, \ldots$ is a sequence of disjoint events whose union is Ω. So Axiom 3 says

$$\mathsf{P}(\Omega) = \mathsf{P}(\Omega) + \mathsf{P}(\emptyset) + \mathsf{P}(\emptyset) + \cdots ,$$

which, along with Axiom 1, requires $\mathsf{P}(\emptyset) = 0$.

The axioms allow $\mathsf{P}(A) = 0$ even if A is not empty.

2. For any *finite* sequence of disjoint events $A_1, \ldots, A_k$,

$$\mathsf{P}(A_1 \cup \cdots \cup A_k) = \sum_{i=1}^{k} \mathsf{P}(A_i). \quad \textit{(finite additivity)}$$

Proof: $A_1, \ldots, A_k, \emptyset, \emptyset, \ldots$ is a sequence of disjoint events whose union is $A_1 \cup \cdots \cup A_k$. Apply Axiom 3 and use the fact $\mathsf{P}(\emptyset) = 0$.

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Properties (cont.)

3. For any event A, $\mathsf{P}(A^c) = 1 - \mathsf{P}(A)$.

Proof: Since A and A^c are disjoint and $A \cup A^c = \Omega$, Property 2 and Axiom 2 say that

$$\mathsf{P}(A) + \mathsf{P}(A^c) = \mathsf{P}(\Omega) = 1.$$

4. If $A \subset B$, then $\mathsf{P}(A) \leq \mathsf{P}(B)$.

Proof: Here, $B = A \cup (B \cap A^c)$, and A and $B \cap A^c$ are disjoint. By Axiom 1, $\mathsf{P}(B \cap A^c) \geq 0$. So, by Property 2,

$$\mathsf{P}(A) = \mathsf{P}(B) - \mathsf{P}(B \cap A^c) \leq \mathsf{P}(B).$$

Careful: it is containment, not "size", that gives this inequality.

5. For any event A, $\mathsf{P}(A) \leq 1$.

Proof: Apply Property 4 with $B = \Omega$.

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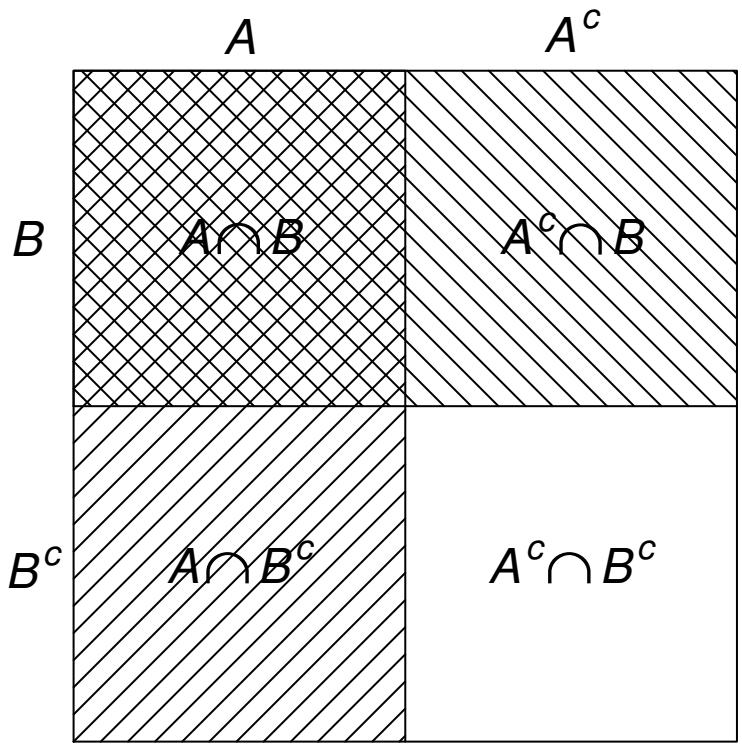
6. For any two events A and B (not necessarily disjoint),

$$\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cap B).$$

Proof: It helps to represent the events in a *Venn-type diagram*:



This shows that $A \cup B = A \cup (A^c \cap B)$, which is a union of disjoint events, implying $\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(A^c \cap B)$.

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Properties (cont.)

Likewise,

$$\mathsf{P}(B) = \mathsf{P}(A \cap B) + \mathsf{P}(A^c \cap B),$$

and so

$$\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(A^c \cap B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cap B).$$

Adding $\mathsf{P}(A)$ and $\mathsf{P}(B)$ accounts for $\mathsf{P}(A \cap B)$ *twice*, which is why it needs to be subtracted. (See the diagram.)

Note: in the special case that A and B are *disjoint*, we have $\mathsf{P}(A \cap B) = \mathsf{P}(\emptyset) = 0$ and thus

$$\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(B),$$

in agreement with Property 2.

This property illustrates a useful point. When looking at the probability of an event, for example one that involves combining other events, it may be helpful to break the event down into disjoint pieces.

Note: an exact expression for $\mathsf{P}(A_1 \cup \cdots \cup A_k)$ *extending Property 6 is somewhat complicated, even with* $k = 3$.

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Facebook and TikTok

Example 1–8. 74% of students at a particular university have a Facebook account, 48% have a TikTok account and 43% have both. What percentage have one or the other?

Technically, there is nothing random here. But, in the context of randomly sampling one student we can treat proportion as the same as probability.

Solution: First, *define the relevant events!* Let $A =$ "student has Facebook" and $B =$ "student has TikTok".

Then $\mathsf{P}(A) = 0.74$, $\mathsf{P}(B) = 0.48$ and $\mathsf{P}(A \cap B) = 0.43$. Hence,

$$\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cap B) = 0.74 + 0.48 - 0.43 = 0.79.$$

What percentage of students have neither account?

Solution: Here we want $\mathsf{P}(A^c \cap B^c)$. We can use DeMorgan's rule: $A^c \cap B^c = (A \cup B)^c$. Thus,

$$\mathsf{P}(A^c \cap B^c) = \mathsf{P}((A \cup B)^c) = 1 - \mathsf{P}(A \cup B) = 1 - 0.79 = 0.21.$$

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Properties (cont.)

7. Suppose $A_1 \subset A_2 \subset A_3 \subset \cdots$ and $A = \bigcup_{n=1}^{\infty} A_n$. Then $\mathsf{P}(A_n) \to \mathsf{P}(A)$ as $n \to \infty$.

Proof: Define $B_1 = A_1$ and $B_n = A_n \cap A_{n-1}^c$ for $n > 1$. Observe that the B_n's are disjoint and $\bigcup_{i=1}^{n} B_i = A_n$. (A picture helps here.) Thus $\mathsf{P}(A_n) = \sum_{i=1}^{n} \mathsf{P}(B_i)$. Furthermore, $\bigcup_{i=1}^{\infty} B_i = A$. So, by the *definition of an infinite sum*,

$$\mathsf{P}(A) = \sum_{i=1}^{\infty} \mathsf{P}(B_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathsf{P}(B_i) = \lim_{n\to\infty} \mathsf{P}(A_n).$$

8. Suppose $A_1 \supset A_2 \supset A_3 \supset \cdots$ and $A = \bigcap_{n=1}^{\infty} A_n$. Then $\mathsf{P}(A_n) \to \mathsf{P}(A)$ as $n \to \infty$.

Proof: Since $A_1^c \subset A_2^c \subset \cdots$, we can just use DeMorgan's Rule and the previous property. (Left for exercise.)

We will use Properties 7 and 8 only occasionally. Theoretically, however, they are extremely important as many interesting events can only be considered by such limits.

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Properties (cont.)

9. *Boole's inequality*. For any events $A_1, \ldots, A_k$,

$$\mathsf{P}\Big(\bigcup_{i=1}^{k} A_i\Big) \leq \sum_{i=1}^{k} \mathsf{P}(A_i).$$

Proof: For two events ($k = 2$) the result is immediate from Property 6 and Axiom 1.

Now suppose the result is true for $k-1$ events. Using both this supposition and the $k = 2$ case, we show that it is true for k events:

$$\begin{aligned}\mathsf{P}\Big(\bigcup_{i=1}^{k} A_i\Big) &= \mathsf{P}\Big((\bigcup_{i=1}^{k-1} A_i)\bigcup A_k\Big) \leq \mathsf{P}\Big(\bigcup_{i=1}^{k-1} A_i\Big) + \mathsf{P}(A_k) \\ &\leq \sum_{i=1}^{k-1} \mathsf{P}(A_i) + \mathsf{P}(A_k) = \sum_{i=1}^{k} \mathsf{P}(A_i).\end{aligned}$$

The conclusion follows for all k by *mathematical induction*.

Recall that mathematical induction needs both (a) a starting case and (b) the property that each new case is implied by the previous ones.

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Properties (cont.)

The properties just discussed are the *tools* with which we can prove results and compute, or at least bound, probabilities. Try to make them second nature and continually ask yourself whether they can be used.

- Always ask if $\mathsf{P}(A^c)$ might be easier to find/compute than $\mathsf{P}(A)$.
- Consider breaking events down into smaller, disjoint pieces that can be managed more easily.
- Even if you eventually want an exact value, using a probability bound might help you to get an idea what the answer is like. Sometimes a bound is all that is needed.
- Never forget to check that your final probability is indeed between 0 and 1. Or between any other bounds you can ascertain.
- If, for example, you are finding $\mathsf{P}(A_n)$ and you know that the events A_n get bigger (each containing its predecessor) as n increases, then so must their probabilities.

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Rolling until a 6 Appears

Example 1–5 (cont.). N = number of rolls of a die until 6 appears. The sample space is $\Omega = \{1, 2, \dots\}$. For an experiment with a fair die, it can be shown that

$$\mathsf{P}(N = n) = \mathsf{P}(\{n\}) = \frac{5^{n-1}}{6^n}.$$

Hence, by Axiom 3,

$$\begin{aligned}\mathsf{P}(N > n) &= \mathsf{P}(\{n+1, n+2, \dots\}) = \sum_{i=n+1}^{\infty} \mathsf{P}(\{i\}) \\ &= \sum_{i=n+1}^{\infty} \frac{5^{i-1}}{6^i} = \frac{1/5(5/6)^{n+1}}{(1-5/6)} = (5/6)^n,\end{aligned}$$

where we have used the *geometric sum rule*: $\sum_{i=n}^{\infty} a^i = \frac{a^n}{1-a}$ when $|a| < 1$.

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Rolling until a 6 Appears

Thus,

$$\begin{aligned} \mathsf{P}(N \le n) &= \mathsf{P}(\{1, \ldots, n\}) = \mathsf{P}(\{n+1, n+2, \ldots\}^c) \\ &= 1 - \mathsf{P}(\{n+1, n+2, \ldots\}) = 1 - (5/6)^n. \end{aligned}$$

Check: The event "$N \le n$" gets bigger as n gets bigger, and so does its probability.

For $1 < m \le n$, the event "$m \le N \le n$" is $\{m, \ldots, n\}$, and

$$\{1, \ldots, n\} = \{1, \ldots, m-1\} \cup \{m, \ldots, n\}.$$

So, by Property 2, $\mathsf{P}(N \le n) = \mathsf{P}(N \le m-1) + \mathsf{P}(m \le N \le n)$, and

$$\mathsf{P}(m \le N \le n) = \mathsf{P}(N \le n) - \mathsf{P}(N \le m-1) = (5/6)^{m-1} - (5/6)^n.$$

Check: This probability is positive (since $m - 1 < n$ and $5/6 < 1$).

Since $\mathsf{P}(N > n) \to 0$ and $\mathsf{P}(N \le n) \to 1$ as $n \to \infty$, there is no chance that a 6 never appears. On the other hand, there are infinitely many possible sequences of rolls which have no 6. This shows that events with 0 probability are not necessarily "trivial".

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1.4 Counting Methods

1.4.1 Finite and Countably Infinite Sample Spaces

If the sample space Ω is countable then without loss we can denote its outcomes by $\omega_1, \omega_2, \ldots$, either finitely many or infinitely many. Then it must be true that

$$\sum_i \mathsf{P}(\{\omega_i\}) = \mathsf{P}(\Omega) = 1.$$

(Note: $\{\omega_i\}$ is an event, which is not quite the same thing as ω_i.)

Any event A can then be represented

$$A = \{\omega_{i_1}, \omega_{i_2}, \ldots\},$$

implying that

$$\mathsf{P}(A) = \textstyle\sum_j \mathsf{P}(\{\omega_{i_j}\}).$$

That is, for countable sample spaces, the probability of any event A is the sum of the probabilities of the outcomes it contains.

Be careful with notation, though. For example, "2" is neither an event nor a random value, so writing $\mathsf{P}(2)$ would not be meaningful.

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Equally Likely Outcomes

If A is a finite set, we will denote its size by $|A|$.

- A finite sample space Ω has *equally likely* outcomes if each outcome ω has probability $\mathsf{P}(\{\omega\}) = \frac{1}{|\Omega|}$.
- If the outcomes in Ω are equally likely, then it follows immediately that

$$\mathsf{P}(A) = \frac{|A|}{|\Omega|}.$$

Therefore, in this case, calculating probabilities merely(!) requires counting the number of outcomes in the relevant events.

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Rolling Two Dice

Example 1–4 (cont.). Consider again our experiment rolling two distinguishable dice. If the dice are balanced ("fair"), then the probability of each of the 36 different outcomes is the same $(1/36)$.

To find the probability of rolling a total of 7, we just need to count the number of outcomes that make up the event. The event "total $= 7$" is

$$\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\},$$

which has six outcomes. Hence,

$$\mathsf{P}(\text{total} = 7) = \frac{6}{36} = \frac{1}{6}\,.$$

Likewise, "total is odd" has 18 outcomes, with probability $\frac{18}{36} = \frac{1}{2}$.

However, if the outcomes are not equally likely then we cannot simply count them to find a probability.

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1.4.2 Counting Rules

Sometimes the sample space has too many outcomes to list conveniently. In such cases, *counting rules* come in handy. The first of these rules is the "mother" of all the others.

- *Product rule*. Suppose that the outcomes of an event can be enumerated in two stages. If there are r_1 different ways for the first stage and, *regardless of those choices*, there are r_2 ways to choose a result for the second stage then the total number of outcomes in the event is $r_1 \times r_2$.
  Note: the actual choices for the second stage can depend on the first stage result – it is the *number* of choices that does not.
- The concept may be extended recursively. Suppose the enumeration takes m stages and, for each k, the number of choices for the k-th stage is r_k, independently of the previous choices, then the event has $r_1 \times \cdots \times r_m$ outcomes.

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Card Example

Example 1–9. The experiment is to draw two cards from a standard deck of 52 cards. How many possible *ordered* hands of two cards are there? (By "ordered" we mean, for example, that $(2\diamondsuit, 5\spadesuit)$ is considered different from $(5\spadesuit, 2\diamondsuit)$.)

On the first draw, there are 52 possibilities. Regardless of which card is drawn, there are 51 possibilities for the second draw. So, the total number of hands is $52(51) = 2652$. This is the size of the sample space.

Now suppose the outcomes are equally likely and consider the event that both cards are spades. On the first draw there are 13 possibilities and regardless of which spade card is drawn there are 12 choices for the second. So the event "both cards are spades" has $13(12) = 156$ outcomes. Therefore,

$$\mathsf{P}(\text{both spades}) = \frac{156}{2652} = \frac{1}{17}.$$

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Card Example (cont.)

If we don't care about the order in which we get the cards, how many hands are there? Does this change the probability?

Answer: Since *every hand corresponds to exactly 2 ordered pairs*, we just divide the total number of outcomes by 2, yielding $2652/2 = 1326$. We do likewise for the event "both spades" and so $\mathsf{P}(\text{both spades}) = \frac{78}{1326} = \frac{1}{17}$, the same value as before.

If accounting for the order (of choices that are made in stages) is relevant, it often can simplify the problem.

Sometimes order is in fact necessary and other times it may be inappropriate. One should always ask whether accounting for order is helpful or relevant.

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Permutations

- *Ordering Rule*. Suppose we have n distinct objects and want to arrange them sequentially. Each possible arrangement of the objects is called a *permutation*. The number of possible permutations of n objects is "n factorial": $n! = n(n-1)(n-2)\cdots(2)1$.

  This is proven using the product rule. There are n stages to choosing which object comes next and the k-th stage has $n+1-k$ choices.

Note: by convention $0! = 1$.

There are $5! = 5(4)(3)(2)1 = 120$ ways to permute five objects $\{1, 2, 3, 4, 5\}$. It would be much too time-consuming to list the possibilities:

$$\begin{array}{c} 1,\ 2,\ 3,\ 4,\ 5 \\ 1,\ 2,\ 3,\ 5,\ 4 \\ 1,\ 2,\ 4,\ 3,\ 5 \\ \vdots \end{array}$$

and then count them one by one.

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- *Permutation Rule*. Suppose we have n distinct objects and we just want to select k of them to arrange sequentially.

  Using the same logic as before, the number of possible results is

$$P_{n,k} = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}\,.$$

  Note that this has k factors, one for each of the k selections.

Example 1–9 (cont.). If we draw 5 cards from a deck of 52, in order, then there are $P_{52,5} = (52)(51)\cdots(48) = 311{,}875{,}200$ possible results.

To find the number of ways the cards could be drawn in a sequence of values (such as $\{$3,4,5,6,7$\}$ or $\{$9,10,J,Q,K$\}$), we observe that there are 9×4 choices for the first (smallest) card and then 4 choices for each of the others. Thus

$$\text{P(cards in a sequence)} = \frac{9(4^5)}{311{,}875{,}200} = 0.00002995.$$

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Example 1–10. What is the probability that at least two people in a group of size n have the same birthday?

Solution: Let us suppose that the 365 birthdays (excluding leap days) are equally likely. There are 365^n possibilities for the vector of n birthdays. But there are $P_{365,n}$ ways to choose a vector of n distinct birthdays. Therefore,

$$\mathsf{P}(\text{all } n \text{ birthdays are different}) = \frac{P_{365,n}}{365^n} = \frac{365!/(365-n)!}{365^n}.$$

The probability p_n that at least two people have the same birthday is

$$p_n = 1 - \mathsf{P}(\text{all } n \text{ birthdays are different}) = 1 - \frac{365!/(365-n)!}{365^n}.$$

n	p_n	n	p_n
10	0.117	25	0.569
15	0.253	30	0.706
20	0.411	35	0.814
21	0.444	40	0.891
22	0.476	45	0.941
23	0.507	50	0.970
24	0.538	60	0.994

Computing such values may take care since, for example, $365!$ is too large for a computer or calculator to handle.

[birthday.r]

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Combinations

- *Combinations Rule*. The number of ways to select k distinct objects from among n, *without regard to order* is denoted $\binom{n}{k}$ (or $C_{n,k}$), and sometimes referred to as "n, choose k".

  Since every unordered selection can be ordered in $k!$ ways, then the number of (ordered) permutations is $P_{n,k} = \binom{n}{k}(k!)$, which we get by using the two-stage product rule. Recalling that $P_{n,k} = \frac{n!}{(n-k)!}$, we thus determine

$$\binom{n}{k} = C_{n,k} = \frac{n!}{k!(n-k)!}.$$

Example 1–9 (cont.). The number of unordered hands of 5 cards is

$$\binom{52}{5} = \frac{52!}{5!\,47!} = 2{,}598{,}960.$$

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The Binomial Coefficient

The combinations number $\binom{n}{k}$ is also called a *binomial coefficient*. For any real numbers x and y, and any nonnegative integer n, the *binomial sum* is

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Note that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \binom{n}{n-k}.$$

Why does this make sense?

Recall that $0!$ is defined to be 1. Some simple cases are

$$\binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = \binom{n}{n-1} = n \quad \text{and} \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

These are useful to memorize.

We will encounter binomial coefficients again when we discuss the binomial distribution.

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Example 1–11. You will select 3 individuals at random from each of the following groups to invite to a party.

Family & Close Friends: 4 men, 5 women.
Clubs & Social Media: 5 men, 3 women.

How many possible configurations are there?

Solution: $\binom{9}{3} \times \binom{8}{3} = 4704$.

What is the chance that all are men?

Solution: $\mathsf{P}(\text{all men}) = \frac{\binom{4}{3}\times\binom{5}{3}}{4704} = \frac{40}{4704} = 0.0085034$.

What is the chance of 2 men from the first group and one from the second?

Solution: Let M_1 and M_2 be the number of men selected from the first and second groups, respectively. *We also have to account for the women selected.*

$$\mathsf{P}(M_1 = 2, M_2 = 1) = \frac{\binom{4}{2}\binom{5}{1} \times \binom{5}{1}\binom{3}{2}}{4704} = \frac{900}{4704} = 0.191327.$$

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Party! (cont.)

What is the chance that you select 3 men total (and hence 3 women)?

Solution: This is more complicated because there are different, unequal, ways that the event can occur. We *split the event into disjoint pieces* according to the number of men selected from each group.

$$
\begin{aligned}
&\mathsf{P}(3 \text{ men and } 3 \text{ women}) \\
&\quad = \mathsf{P}(M_1 = 0, M_2 = 3) + \mathsf{P}(M_1 = 1, M_2 = 2) + \mathsf{P}(M_1 = 2, M_2 = 1) \\
&\qquad + \mathsf{P}(M_1 = 3, M_2 = 0) \\
&\quad = \frac{\binom{4}{0}\binom{5}{3} \times \binom{5}{3}\binom{3}{0} + \binom{4}{1}\binom{5}{2} \times \binom{5}{2}\binom{3}{1} + \binom{4}{2}\binom{5}{1} \times \binom{5}{1}\binom{3}{2} + \binom{4}{3}\binom{5}{0} \times \binom{5}{0}\binom{3}{3}}{4704} \\
&\quad = \frac{1754}{4704} = 0.372874.
\end{aligned}
$$

Always consider whether expressing an event as a union of disjoint pieces might simplify the computation.

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1.4.3 Simple Random Sample

Consider a population consisting of N distinct elements. There are two common, and distinct, ways to draw a sample from the population.

- *Sampling With Replacement.* Suppose a sample of n elements is drawn sequentially in such a way that the element drawn on the $(k-1)$-th draw is *replaced* before the k-th element is drawn, $k = 2, \ldots, n$. The number of possible samples is:

$$N \times N \times \cdots \times N = N^n.$$

  It is necessary to treat the sample as ordered in this case.

- *Sampling Without Replacement.* A sample of n elements ($n \leq N$) is drawn sequentially *without replacing* any elements along the way. The number of possible samples is:

$$N(N-1)(N-2)\cdots(N-n+1) = \frac{N!}{(N-n)!}.$$

  If order does not matter then there are $\binom{N}{n}$ possible unordered samples.

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Facebook Friends

Example 1–12. A simple random sample of size $n = 4$ is drawn without replacement from your N Facebook friends ($N \geq 4$). What is the probability that friends Jack, Susan and Harry are in the sample?

First of all, there are $\frac{N!}{(N-4)!}$ outcomes for the experiment and they are equally likely since the selections are made at random.

How many possible samples of size $n = 4$ contain Jack, Susan and Harry?

$$4! \times (1)(1)(1)(N-3) = 24(N-3).$$

Again, all of the samples are equally likely. So the desired probability is

$$\frac{24(N-3)}{N!/(N-4)!} = \frac{24(N-3)!}{N!} = \frac{24}{N(N-1)(N-2)}.$$

Since the neither the sample nor the event depend on order, we can also calculate the probability by

$$\frac{N-3}{\binom{N}{4}} = \frac{24}{N(N-1)(N-2)}.$$

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